Face2Health: A Decoupled Multimodal Machine Learning Framework for Screening Physician-Diagnosed Diabetes and Hypertension Risk

Phalathip Hemvut
School of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
67070255@it.kmitl.ac.th

Eua-Unggul Chaivivatporn
School of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
67070302@it.kmitl.ac.th

***Abstract*—This paper presents Face2Health, a decoupled multimodal machine learning architecture engineered for accessible screening of physician-diagnosed Type 2 Diabetes Mellitus (DIQ010) and Essential Hypertension (BPQ020). The pipeline sequentially couples: (1) a Vision Transformer (ViT-H/14) fine-tuned on VisualBMI and Chicago Face Database (CFD) for Body Mass Index (BMI) estimation (MAE = 2.14 kg/m², RMSE = 2.82 kg/m², R^2 = 0.68) with 25-pass Monte Carlo Dropout for epistemic uncertainty rejection (σ > 1.80 kg/m², 4.2% rejection rate) alongside scale-invariant facial morphometrics; (2) a tabular Extreme Gradient Boosting (XGBoost) regressor mapping estimated BMI, age, biological sex, and physical activity level to Dual-Energy X-ray Absorptiometry (DEXA)-derived Total Body Fat Percentage (R^2 = 0.74, MAE = 3.42% for non-invasive route; R^2 = 0.82, MAE = 2.85% with waist circumference); and (3) monotonically regularized, Platt-calibrated XGBoost classifiers estimating continuous posterior probabilities. Evaluated on the CDC NHANES adult cohort (2011–2018 cycles, aged 18–59 years, N = 3,540 for Diabetes, N = 3,554 for Hypertension) using out-of-fold cross-validation and noisy vision error simulation, the primary non-invasive pipeline (without waist circumference) achieves 90.91% sensitivity for Diabetes (50/55 positive cases detected, 95% CI: 80.0%–96.7%) and 90.61% sensitivity for Hypertension (164/181 positive cases detected, 95% CI: 85.3%–94.4%) under F2-optimized screening thresholds (τ = 0.060 for Diabetes, τ = 0.155 for Hypertension). We present comprehensive baseline comparisons against Logistic Regression, execute full ablation studies, detail participant inclusion flow, and establish an epidemiologically aligned 4-tier clinical triage protocol.**

***Keywords*—computer vision, vision transformer, MC dropout, non-invasive health screening, class imbalance, probability calibration, Platt scaling, XGBoost, monotonicity constraints, diabetes, hypertension, NHANES, DEXA.**

I. INTRODUCTION & PROBLEM STATEMENT

Non-communicable diseases (NCDs), led by Type 2 Diabetes Mellitus and Essential Hypertension, constitute the defining epidemiological crisis of the modern era, accounting for over 41 million deaths annually—equivalent to 74% of all mortalities worldwide [1]. Beyond mortality, sustained undetected hyperglycemia and arterial hypertension induce progressive microvascular and macrovascular pathology, culminating in diabetic nephropathy, proliferative retinopathy, peripheral neuropathy, ischemic stroke, and coronary heart disease. In middle-income and newly industrialized nations, population health surveys indicate that 30% to 45% of individuals living with diabetes and hypertension remain undiagnosed until secondary clinical complications necessitate emergency hospitalization [2].

Current gold-standard diagnostic pathways rely on invasive venous phlebotomy (fasting plasma glucose ≥ 126 mg/dL or HbA1c ≥ 6.5%) and clinical sphygmomanometry. While biochemically authoritative, these methods impose structural limitations requiring sterile consumables, certified medical technologists, and centralized laboratory infrastructure. In parallel, clinical anthropometry demonstrates that systemic adiposity and metabolic risk physically manifest in craniofacial soft tissue. Adipocyte hypertrophy within the superficial and deep buccal fat compartments, jowl fat expansion, and submental tissue thickening reflect central lipid deposition. The maturation of Vision Transformers (ViT) with self-attention, alongside gradient boosted decision trees (XGBoost), provides a computational framework capable of translating passive 2D digital portraits into physiological biomarkers.

However, cascading visual-to-tabular pipelines introduce three key algorithmic challenges: (1) multi-stage regression error propagation, where vision estimation noise affects subsequent metabolic models; (2) biological inconsistency, where collinear features lead unconstrained decision trees to form non-monotonic risk surfaces; and (3) probability output compression under class imbalance (~10.3% diabetes prevalence), where standard 0.50 decision thresholds yield 0.0% default clinical sensitivity. This paper formalizes, benchmarks, and resolves these operational dynamics on the CDC NHANES dataset using physician-diagnosed disease targets (DIQ010 and BPQ020).

II. LITERATURE REVIEW & BENCHMARK ANALYSIS

A. Craniofacial Morphometry and Visceral Adiposity

Craniofacial morphology serves as an external indicator of systemic adiposity due to the distribution of subcutaneous and deep facial fat pads. Landmark anatomical studies by Coetzee et al. [3] established that facial adiposity serves as a reliable cue for systemic health. L. Wen and G. Guo [4] demonstrated that computer-derived facial features correlate significantly with body mass index and metabolic parameters. Anatomically, lipid accumulation induces lateral and inferior displacement across the mandibular border. To capture these shifts, four scale-invariant geometric indices are defined: Lower Facial Width-to-Height Ratio (LFWR), Cheek-to-Jaw Width Ratio (CJWR), Perimeter-to-Area Ratio (PAR), and Facial Width-to-Height Ratio (FWHR). Lee and Kim [5] demonstrated that facial width measurements correlate significantly with abdominal obesity metrics.

B. Facial Biomarker Estimation & Deep Learning Benchmarks

Early facial health estimation relied on social media datasets. Kocabey et al. [6] introduced *Face-to-BMI* using public Reddit progress photographs, establishing initial CNN benchmarks (MAE ~2.8 kg/m²). Stephen et al. [7] demonstrated cross-cultural validity of facial shape cues across Caucasian, African, and Asian cohorts. Jiang, Guo, and Mu [8] extended facial adiposity models to metabolic risk prediction using deep residual networks [19]. However, these monolithic approaches map pixels directly to disease labels, lacking epistemic uncertainty estimation and intermediate biomarker transparency.

C. Dual-Energy X-ray Absorptiometry (DEXA) & NHANES Cohort

While Body Mass Index (BMI = weight/height²) is ubiquitous, it cannot differentiate skeletal muscle mass from adipose tissue. Dual-Energy X-ray Absorptiometry (DEXA) constitutes the diagnostic gold standard for body composition analysis, using photon attenuation to

resolve bone mineral, lean soft tissue, and fat mass. The National Health and Nutrition Examination Survey (NHANES) conducted by the CDC represents a multi-ethnic dataset combining whole-body DEXA scans (total body fat percentage DXDTOPF), standardized anthropometry (BMX), laboratory biochemical assays (LBXGLU, LBXGH), and structured diagnostic interviews (DIQ010 for physician-diagnosed diabetes, BPQ020 for physician-diagnosed hypertension) across continuous 2011–2018 survey cycles for adult participants aged 18–59 years [16].

D. Vision Transformers and Epistemic Uncertainty Quantification

Dosovitskiy et al. [9] showed that Vision Transformers (ViT) model long-range spatial dependencies across distant facial landmarks using self-attention. Deterministic neural networks, however, suffer from overconfident mispredictions on out-of-distribution inputs. To address this, Monte Carlo Dropout (Gal & Ghahramani [10]) provides a Bayesian approximation of Gaussian process uncertainty by preserving active dropout masks during inference. Sampling T=25 forward stochastic passes yields the predictive mean $\mu(x^*)$ and epistemic variance $\sigma^2(x^*)$:

$$\mu(x^*) = (1/T)\,\Sigma_{t=1}{}^{T} f_{W_t}(x^*), \qquad \sigma^2(x^*) = [1/(T-1)]\,\Sigma_{t=1}{}^{T} [f_{W_t}(x^*) - \mu(x^*)]^2$$

If $\sigma(x^*)$ exceeds 1.80 kg/m², the system rejects the input as an epistemic anomaly.

E. Monotonic Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) [11] is an optimized distributed gradient boosted decision tree algorithm [18]. Standard greedy tree construction can produce erratic step functions when handling collinear predictors. In clinical medicine, risk must adhere to physiological gradients (e.g., escalating age, BMI, or body fat cannot biologically decrease diabetes risk). XGBoost allows the enforcement of monotonicity constraints ($c_j = +1$) on Age, BMI, Body Fat %, and Waist Circumference, while biological sex remains unconstrained ($c_j = 0$).

F. Platt Scaling & Probability Calibration

Uncalibrated machine learning models output arbitrary ordinal scores. In epidemiological triage, output scores should represent true posterior probabilities. Platt Scaling [12] fits a sigmoid logistic function over validation margins z using maximum likelihood log-loss estimation: $P(Y=1|z) = 1 / [1 + \exp(A \cdot z + B)]$. Calibration quality is commonly assessed with the Brier score [13]; calibrated posterior probabilities naturally reflect the empirical prevalence of the training cohort (~10.3% for diabetes), resulting in compressed probability ranges where standard 0.50 cutoffs fail.

G. Decision Theory & Threshold Optimization

In clinical disease screening, classification error costs are heavily asymmetric. The health cost of a False Negative (delayed diagnosis) exceeds that of a False Positive (requiring a low-cost blood or blood pressure re-test) [15]. Decision theory dictates adjusting classification thresholds τ below 0.50 [14]. We evaluate Youden's J Statistic ($J(\tau)$ = Sensitivity + Specificity - 1) and the F_β measure (β=2, weighting Recall twice as heavily as Precision) to optimize screening sensitivity.

III. METHODOLOGY & SYSTEM ARCHITECTURE

A. Process Overview & Cascading Architecture

The Face2Health framework executes across three sequential, modular stages:

1) Stage 1 (Computer Vision & Morphometry): A frontal portrait is preprocessed via MediaPipe Face Mesh. If face pose exceeds ±15° in pitch, yaw, or roll, the Face Guard module rejects the frame. Validated faces are processed by ViT-H/14 (fine-tuned on VisualBMI and Chicago Face Database). MC Dropout (25 passes) produces estimated BMI μ and epistemic uncertainty σ. Simultaneously, landmark coordinates yield LFWR, CJWR, PAR, and FWHR ratios as supplementary morphometric features.

2) Stage 1.5 (Tabular DEXA Regression): Estimated BMI, age, biological sex (male=1, female=0), and physical activity level are passed to an XGBoost regressor trained on NHANES DEXA scans (N=2,477 training split using 5-fold out-of-fold predictions and hyperparameter optimization strategies [20]), achieving $R^2 = 0.74$ and MAE = 3.42% body fat for the primary non-invasive route.

3) Stage 2 (Monotonic Calibrated Risk Classification): Tabular features [Age, Sex, Predicted BMI, Predicted Body Fat, (Optional Waist)] are evaluated by monotonically regularized XGBoost classifiers calibrated via 5-fold internal Platt scaling [17] for physician-diagnosed Diabetes and Hypertension.

TABLE I. CDC NHANES ADULT COHORT PARTICIPANT FLOW & PARTITIONING (AGED 18–59 YEARS)

Filtering Step	Criteria / Description	Remaining (N)	Positive Cases	Base Prev.
Initial NHANES Cohort (2011–2018)	Continuous 4-cycle survey participants	39,156	—	—
Age Filter (Adult Cohort)	Participants aged 18–59 years (DXA eligible)	14,210	—	—
DEXA Examination Protocol	Valid whole-body DXA scan (DXDTOPF), <450 lbs, <6'5"	5,120	—	—
Diabetes Cohort (DIQ010 Target)	Complete diagnostic questionnaire responses	**3,540**	**366**	**10.34%**
— Train Split (70%)	Out-of-fold training partition	2,477	256	10.34%
— Validation Split (15%)	Threshold optimization partition (τ)	532	55	10.34%
— Held-Out Test Split (15%)	Unobserved final evaluation partition	531	55	10.36%
Hypertension Cohort (BPQ020 Target)	Complete diagnostic questionnaire responses	**3,554**	**1,207**	**33.96%**
— Train Split (70%)	Out-of-fold training partition	2,488	845	33.96%
— Validation Split (15%)	Threshold optimization partition (τ)	533	181	33.96%
— Held-Out Test Split (15%)	Unobserved final evaluation partition	533	181	33.96%

B. Decoupled Evaluation & Out-of-Fold Protocol

To ensure academic rigor, Stage 1.5 DEXA regressors were trained using 5-fold cross-validation on the 70% training split. Out-of-fold predictions were generated for training Stage 2 classifiers, eliminating intra-cohort data leakage. During Stage 2 test evaluation, Gaussian noise matching empirical ViT estimation error (σ = 2.14 kg/m²) was injected into test BMI inputs to simulate real-world computer vision propagation.

C. Sensitivity and Error Propagation Analysis

Let predicted body fat θ be a function of estimated BMI b, age a, sex s, and waist w: $\theta = f(b, a, s, w)$. Empirical perturbation analysis on the test cohort indicates that when waist circumference is present, waist features absorb split variance (68.42% feature split importance), yielding an empirical error attenuation coefficient of $\alpha = 0.39$ ($\Delta BF \approx 0.39 \cdot \Delta BMI$). In the primary non-invasive model (without waist), biological sex commands 74.99% feature split importance, while vision estimation error propagates directly with $\alpha = 1.00$ ($\Delta BF \approx 1.00 \cdot \Delta BMI$).

IV. EXPERIMENTAL RESULTS AND COMPARATIVE BENCHMARKS

A. Stage 1 ViT & Stage 1.5 DEXA Regression Performance

Table II summarizes performance metrics across Stage 1 ViT vision estimation and Stage 1.5 DEXA body fat regression models evaluated on held-out test splits.

TABLE II. STAGE 1 VISION AND STAGE 1.5 DEXA REGRESSION BENCHMARKS

Model	Target	Input Features	MAE	RMSE	R^2	Rejection
Stage 1 ViT-H/14 + MC Dropout	BMI (kg/m²)	2D Facial Image	2.14 kg/m²	2.82 kg/m²	0.68	4.2% (σ>1.8)
Stage 1.5 DEXA Regressor (Primary)	Body Fat %	BMI, Age, Sex, Physical Activity	3.42%	4.25%	0.74	0.0%
Stage 1.5 DEXA Regressor (Auxiliary)	Body Fat %	BMI, Age, Sex, Waist	2.85%	3.62%	0.82	0.0%

B. Stage 2 Classification Benchmarks & Baseline Comparisons

Evaluating the baseline Stage 2 diabetes classifier on the held-out test set (N=531) revealed predicted probabilities bounded by max(p) = 0.3776 due to low disease prevalence (~10.3%) and moderate discriminative capability (AUC = 0.762). Because no individual crossed the default 0.50 cutoff, the default model exhibited 0.00% sensitivity (55/55 false negatives). Shifting to the validation-optimized F2 screening threshold (τ = 0.060) on the primary non-invasive model (No Waist) achieved **90.91% sensitivity** (50/55 positive cases detected, 95% CI: 80.0%–96.7%). When waist circumference is optionally included, the F2 threshold (τ = 0.061) achieved **89.09% sensitivity** (49/55 cases detected, 95% CI: 77.8%–95.9%).

TABLE III-A. DIABETES MELLITUS CLASSIFICATION BENCHMARK (N=531, HELD-OUT TEST; VALUES IN PARENTHESES INDICATE 95% CI)

Model	Rule	Cutoff (τ)	AUC	Sens.	Spec.	Prec.
Logistic Reg.	Default	0.500	0.742 (0.678–0.806)	0.0% (0.0–6.5%)	100.0% (99.2–100%)	0.0%
Logistic Reg.	Opt. (F2)	0.104	0.742 (0.678–0.806)	74.5% (61.0–85.3%)	61.3% (56.8–65.7%)	18.2% (13.7–23.5%)
XGBoost	Default	0.500	0.747 (0.683–0.811)	0.0% (0.0–6.5%)	100.0% (99.2–100%)	0.0%
Primary (Ours)	Default	0.500	0.762 (0.701–0.823)	0.0% (0.0–6.5%)	100.0% (99.2–100%)	0.0%
Primary (Ours)	Youden's J	0.082	0.762 (0.701–0.823)	78.2% (65.0–88.2%)	59.0% (54.5–63.5%)	18.1% (13.4–23.6%)
Primary (Ours)	**Opt. (F2)**	**0.060**	**0.762 (0.701–0.823)**	**90.9% (80.0–96.7%)**	**48.7% (44.1–53.3%)**	**17.1% (13.0–21.9%)**
Auxiliary With Waist	Default	0.500	0.763 (0.702–0.824)	0.0% (0.0–6.5%)	100.0% (99.2–100%)	0.0%
Auxiliary With Waist	Youden's J	0.082	0.763 (0.702–0.824)	78.2% (65.0–88.2%)	58.8% (54.3–63.3%)	18.0% (13.3–23.5%)
Auxiliary With Waist	Opt. (F2)	0.061	0.763 (0.702–0.824)	89.1% (77.8–95.9%)	49.0% (44.4–53.6%)	16.8% (12.7–21.6%)

TABLE III-B. ESSENTIAL HYPERTENSION CLASSIFICATION BENCHMARK (N=533, HELD-OUT TEST; VALUES IN PARENTHESES INDICATE 95% CI)

Model Architecture	Rule	Cutoff (τ)	AUC	Sens.	Spec.	Prec.
Logistic Reg.	Default	0.500	0.740 (0.697–0.783)	48.1% (40.6–55.6%)	83.5% (79.2–87.2%)	60.0% (51.9–67.7%)
Logistic Reg.	Opt. (F2)	0.280	0.740 (0.697–0.783)	68.0% (60.6–74.7%)	65.6% (60.4–70.6%)	50.4% (43.8–57.0%)
Unconstrained XGB	Default	0.500	0.750 (0.708–0.792)	49.2% (41.7–56.7%)	82.1% (77.7–85.9%)	58.6% (50.5–66.3%)
Primary No Waist (Ours)	Default	0.500	0.757 (0.715–0.799)	50.3% (42.8–57.8%)	82.7% (78.3–86.5%)	59.9% (51.8–67.5%)
Primary No Waist (Ours)	Youden's J	0.339	0.757 (0.715–0.799)	71.3% (64.1–77.7%)	67.0% (61.8–72.0%)	52.6% (46.2–59.0%)
Primary No Waist (Ours)	**Opt. (F2)**	**0.155**	**0.757 (0.715–0.799)**	**90.6% (85.3–94.4%)**	**40.9% (35.7–46.3%)**	**44.1% (39.0–49.3%)**
Auxiliary With Waist	Default	0.500	0.755 (0.713–0.797)	49.7% (42.2–57.2%)	82.4% (78.0–86.2%)	59.2% (51.1–66.9%)
Auxiliary With Waist	Opt. (F2)	0.158	0.755 (0.713–0.797)	88.4% (82.8–92.6%)	41.8% (36.6–47.1%)	44.0% (38.8–49.2%)

C. Component Ablation Study

To quantify individual contribution across pipeline components, systematic ablation experiments were executed on the Diabetes test cohort (N=531), summarizing AUC, Sensitivity, Specificity, Brier score, and Monotonicity violations in Table IV.

TABLE IV. COMPONENT ABLATION STUDY ON DIABETES TEST COHORT (N=531)

Configuration	AUC	Sens (F2)	Spec (F2)	Brier	Violations
Full Face2Health Pipeline (Ours)	**0.762**	**90.9%**	48.7%	**0.078**	**0 (Guaranteed)**
No Body Fat	0.748	83.6%	51.2%	0.085	0 Violations
No Monotonicity	0.747	85.5%	50.0%	0.084	14 Violations
No Calibration	0.747	80.0%	52.5%	0.142	14 Violations

D. Body Fat Error Attenuation Dynamics

Table V evaluates total body fat shifts under systematic BMI perturbation sweeps (ΔBMI $\in$ [-2, +2] kg/m²), demonstrating the variance shock absorber effect under auxiliary waist features.

TABLE V. BODY FAT SHIFT UNDER BMI PERTURBATION SWEEPS

Injected ΔBMI	Mean ΔBF Waist	Quotient (α)	Mean ΔBF No Waist
-2.0 kg/m²	-0.80 ± 0.80%	0.40	-2.02 ± 1.24%
-1.0 kg/m²	-0.37 ± 0.50%	0.37	-1.01 ± 0.82%
0.0 kg/m²	0.00 ± 0.00%	—	0.00 ± 0.00%
+1.0 kg/m²	+0.39 ± 0.54%	0.39	+1.00 ± 0.83%
+2.0 kg/m²	+0.77 ± 0.72%	0.38	+1.98 ± 1.14%

V. CLINICAL TRIAGE PROTOCOL & RISK SENSITIVITY ANALYSIS

The calibrated posterior probabilities are operationalized into four clinical triage strata configured in weights/thresholds.json. Table VI details population breakdown and observed true disease prevalence across held-out test splits.

TABLE VI. EPIDEMIOLOGICALLY ALIGNED 4-TIER CLINICAL TRIAGE STRATA METRICS

Triage Tier	Probability Range	Diabetes N	Diabetes Prev.	Hypertension Count N	Hypertension Prev.	Action Protocol
Low Risk	p < 6.0% (Dia) p < 15.5% (HTN)	232 (43.7%)	2.15% (5/232)	161 (30.2%)	10.56% (17/161)	Annual wellness checkup
Watchful	6.0% ≤ p < 10.0% 15.5% ≤ p < 25.0%	132 (24.9%)	7.58% (10/132)	122 (22.9%)	27.05% (33/122)	Lifestyle counseling & re-test
Screen Positive	10.0% ≤ p < 20.0% 25.0% ≤ p < 50.0%	121 (22.8%)	18.18% (22/121)	150 (28.1%)	46.67% (70/150)	Actionable phlebotomy / BP trigger
Urgent	p ≥ 20.0% (Dia) p ≥ 50.0% (HTN)	46 (8.7%)	**39.13% (18/46)**	100 (18.8%)	**61.00% (61/100)**	Priority medical referral

VI. LIMITATIONS, PRIVACY, AND ETHICAL CONSIDERATIONS

Face2Health is strictly an opportunistic triage tool, not a diagnostic instrument. Key limitations include: (1) target definitions rely on self-reported physician diagnoses (DIQ010/BPQ020) rather than active prospective laboratory testing; (2) potential domain shift between the US NHANES multi-ethnic training cohort and Asian/Southeast Asian populations regarding BMI-body fat relationships; (3) sensitivity to severe lighting extremes and head poses exceeding ±15° (intercepted by Face Guard); and (4) decoupled pipeline validation due to the absence of paired facial photographs in NHANES. Biometric data handling is designed to comply with PDPA and GDPR guidelines, executing feature extraction locally on-device without storing raw facial images.

VII. CONCLUSION & FUTURE WORK

This study evaluated **Face2Health**, demonstrating that passive computer vision can be integrated into a calibrated triage instrument. By optimizing decision thresholds on validation partitions, the primary non-invasive pipeline (without waist) achieved 90.91% sensitivity for Diabetes and 90.61% for Hypertension on held-out NHANES test data. Future work focuses on: (1) prospective clinical validation in Southeast Asian outpatient clinics; (2) integrating remote photoplethysmography (rPPG) for optical pulse wave velocity estimation; and (3) deploying lightweight ONNX models for offline mobile triage.

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